

Packet Analysis with Wireshark

Analysis on Network Packets using Wireshark

Aayush Dhakal

Bachelors in Computer Networking and IT Security

Islington College, Nepal

August, 2026

1 Context Overview

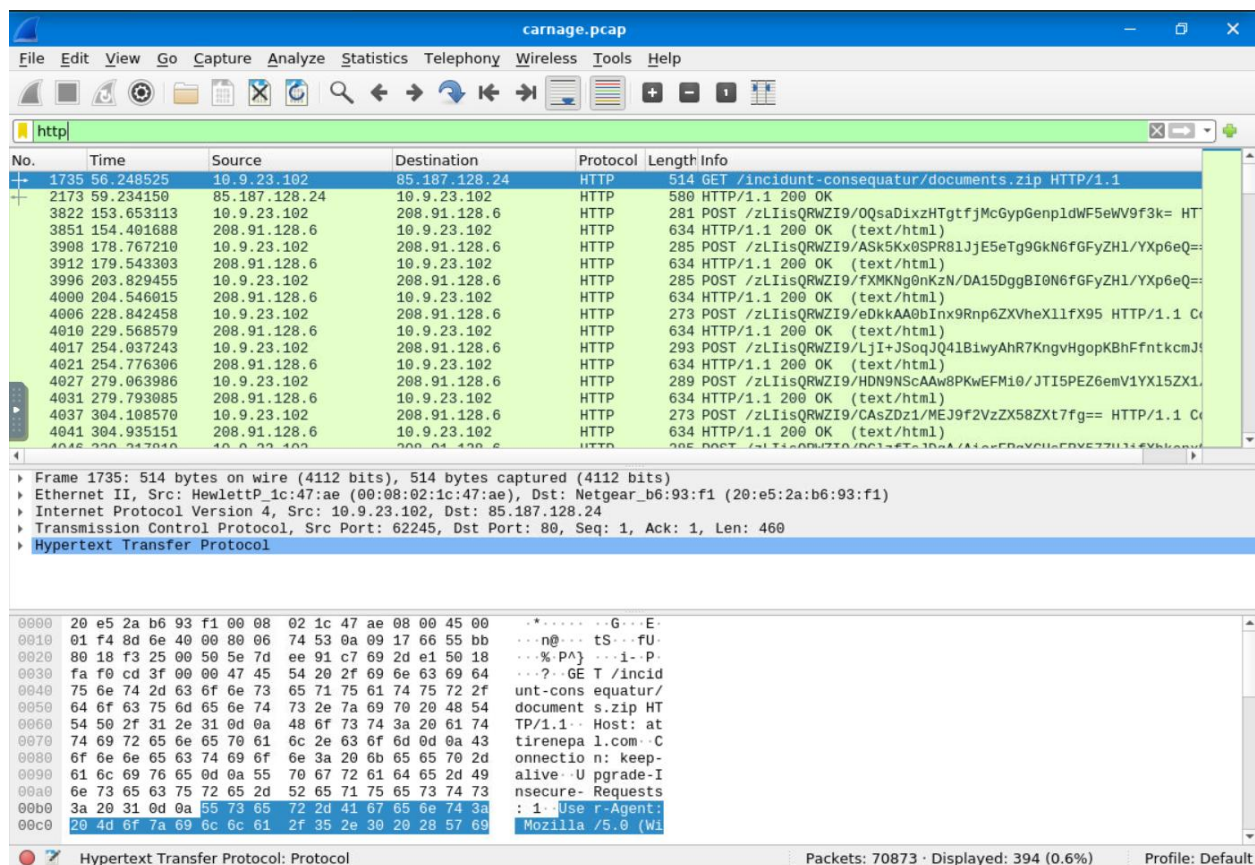
Background: This is an analysis of packet capture file from Carnage room in Wireshark.

Scenario: Eric Fischer from the Purchasing Department at Bartell Ltd has received an email from a known contact with a Word document attachment. Upon opening the document, he accidentally clicked on "Enable Content." The SOC Department immediately received an alert from the endpoint agent that Eric's workstation was making suspicious connections outbound. The pcap was retrieved from the network sensor and handed for analysis.

2 Analysis

2.1 File analysis

A zip file was downloaded by the client. By filtering with 'http', we get a packet with documents.zip file.



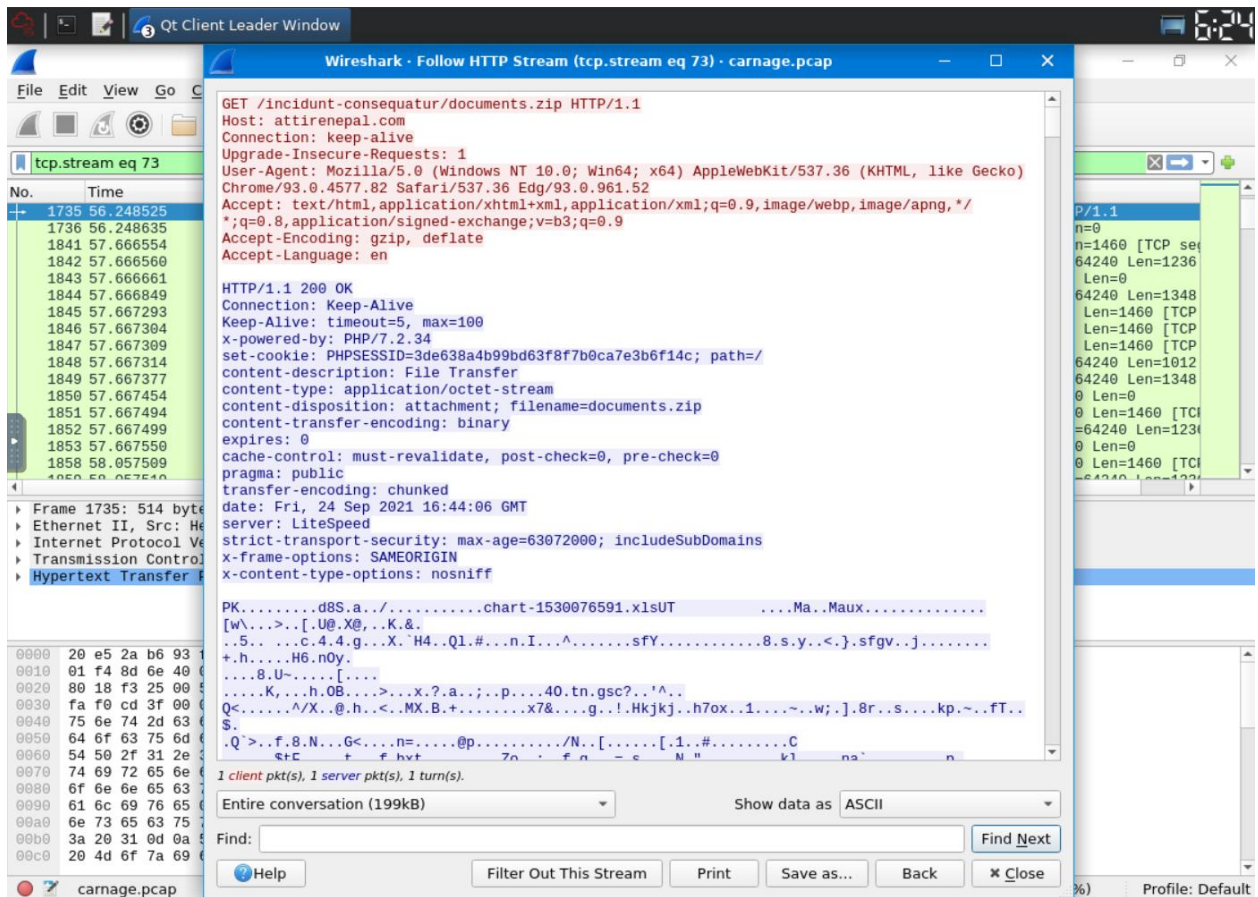
The image shows a Wireshark network traffic capture titled 'carnage.pcap'. The filter bar is set to 'http'. The packet list shows a GET request for '/incidunt-consequatur/documents.zip' from 10.9.23.102 to 85.187.128.24. The packet details pane shows the following structure:

- Frame 1735: 514 bytes on wire (4112 bits), 514 bytes captured (4112 bits)
- Ethernet II, Src: HewlettP_1c:47:ae (00:08:02:1c:47:ae), Dst: Netgear_b6:93:f1 (20:e5:2a:b6:93:f1)
- Internet Protocol Version 4, Src: 10.9.23.102, Dst: 85.187.128.24
- Transmission Control Protocol, Src Port: 62245, Dst Port: 80, Seq: 1, Ack: 1, Len: 460
- Hypertext Transfer Protocol

The raw packet data (hex and ASCII) is shown at the bottom:

```
0000 20 e5 2a b6 93 f1 00 08 02 1c 47 ae 08 00 45 00  ..*...G...E
0010 01 f4 8d 6e 40 00 80 06 74 53 0a 09 17 66 55 bb  ..n@...tS...fu
0020 80 18 f3 25 00 50 5e 7d ee 91 c7 69 2d e1 50 18  ...%PA) ...i-P
0030 fa f0 cd 3f 00 00 47 45 54 20 2f 69 6e 63 69 64  ...?..GE T /incid
0040 75 6e 74 2d 63 6f 6e 73 65 71 75 61 74 75 72 2f  unt-cons equatur/
0050 64 6f 63 75 6d 65 6e 74 73 2e 7a 69 70 20 48 54  document s.zip HT
0060 54 50 2f 31 2e 31 0d 0a 48 6f 73 74 3a 20 61 74  TP/1.1.. Host: at
0070 74 69 72 65 6e 65 70 61 6c 2e 63 6f 6d 0d 0a 43  tirenepa l.com .C
0080 6f 6e 6e 65 63 74 69 6f 6e 3a 20 6b 65 65 70 2d  onnectio n: keep-
0090 61 6c 69 76 65 0d 0a 55 70 67 72 61 64 65 2d 49  alive..U pgrade-I
00a0 6e 73 65 63 75 72 65 2d 52 65 71 75 65 73 74 73  nsecure- Requests
00b0 3a 20 31 0d 0a 55 73 65 72 2d 41 67 65 6e 74 3a  : 1..Use r-Agent:
00c0 20 4d 6f 7a 69 6c 6c 61 2f 35 2e 30 20 28 57 69  Mozilla /5.0 (Wi
```

Since it uses unencrypted HTTP protocol, various information can be found. In the HTTP stream, we can get the host, webserver information, cookie and many more. In the actual payload, we can also see encrypted text, and there is 'chart-1530076591.xls' which is the name of the file that is contained in the documents.zip file.



Host: attirenepal.com

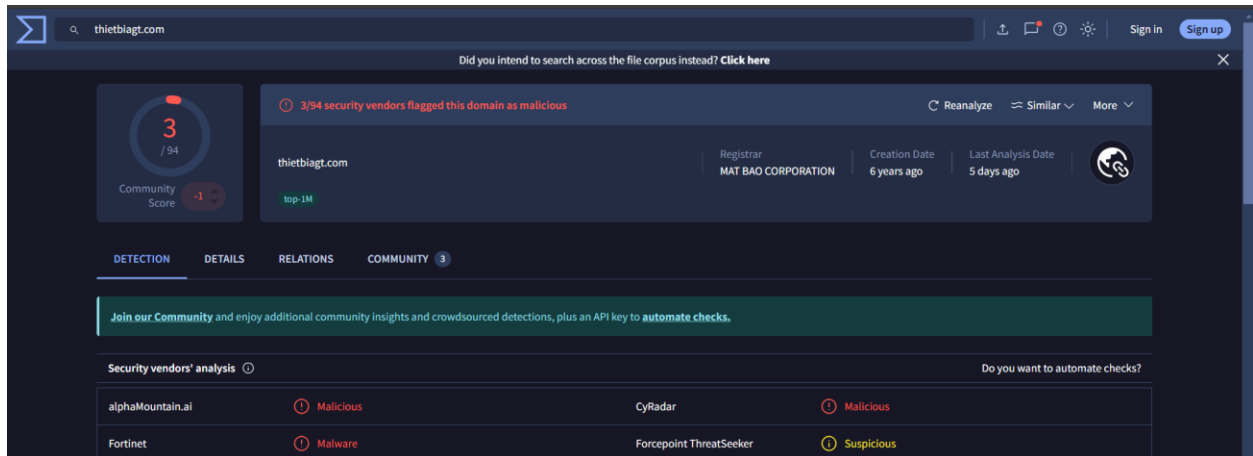
Server: LiteSpeed

Server version (x-powered-by): PHP/7.2.34

File contained: chart-1530076591.xls

2.2 Domain analysis

Server name was added to column and the names were inspected. From the names, three suspicious looking domains were thietbiagt.com, new.americold.com, finejewels.com.au. These domains were checked in VirusTotal.



Did you intend to search across the file corpus instead? [Click here](#)

Community Score: 3 / 94

3/94 security vendors flagged this domain as malicious

thietbiagt.com

Registrar: MAT BAO CORPORATION | Creation Date: 6 years ago | Last Analysis Date: 5 days ago

top-1M

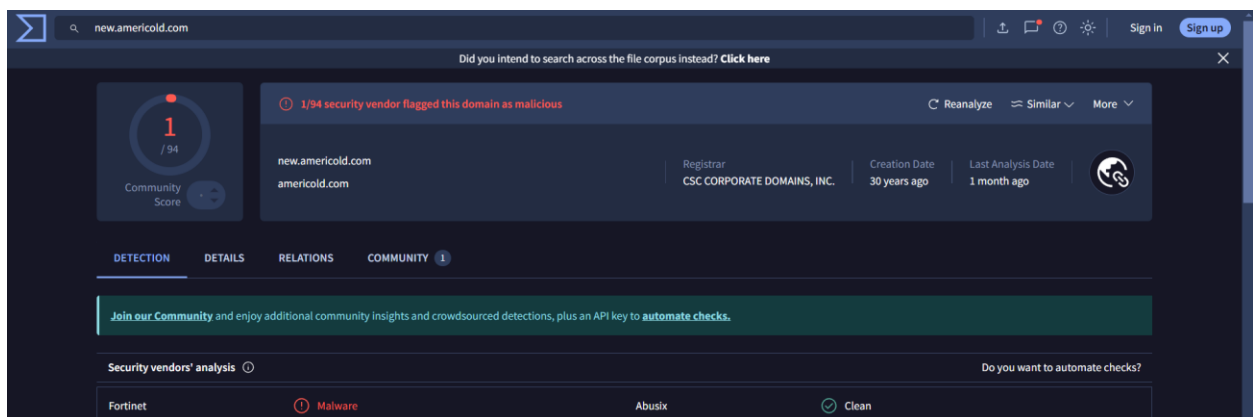
DETECTION DETAILS RELATIONS COMMUNITY 3

Join our Community and enjoy additional community insights and crowdsourced detections, plus an API key to [automate checks](#).

Security vendors' analysis

alphaMountain.ai	Malicious	CyRadar	Malicious
Fortinet	Malware	Forcepoint ThreatSeeker	Suspicious

Do you want to automate checks?



Did you intend to search across the file corpus instead? [Click here](#)

Community Score: 1 / 94

1/94 security vendor flagged this domain as malicious

new.americold.com

Registrar: CSC CORPORATE DOMAINS, INC. | Creation Date: 30 years ago | Last Analysis Date: 1 month ago

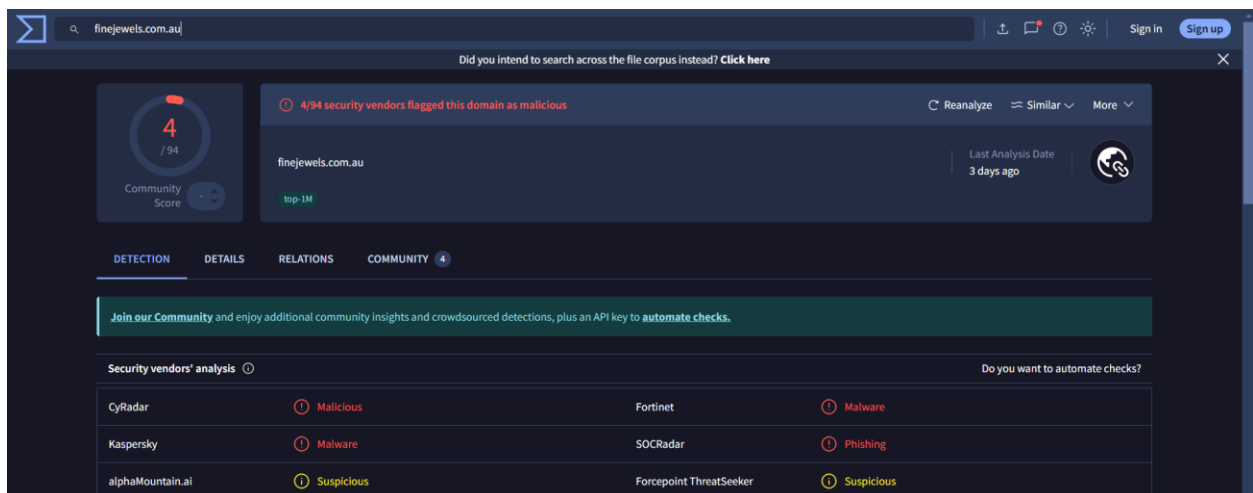
DETECTION DETAILS RELATIONS COMMUNITY 1

Join our Community and enjoy additional community insights and crowdsourced detections, plus an API key to [automate checks](#).

Security vendors' analysis

Fortinet	Malware	Abusix	Clean
----------	---------	--------	-------

Do you want to automate checks?



Did you intend to search across the file corpus instead? [Click here](#)

Community Score: 4 / 94

4/94 security vendors flagged this domain as malicious

finejewels.com.au

Last Analysis Date: 3 days ago

DETECTION DETAILS RELATIONS COMMUNITY 4

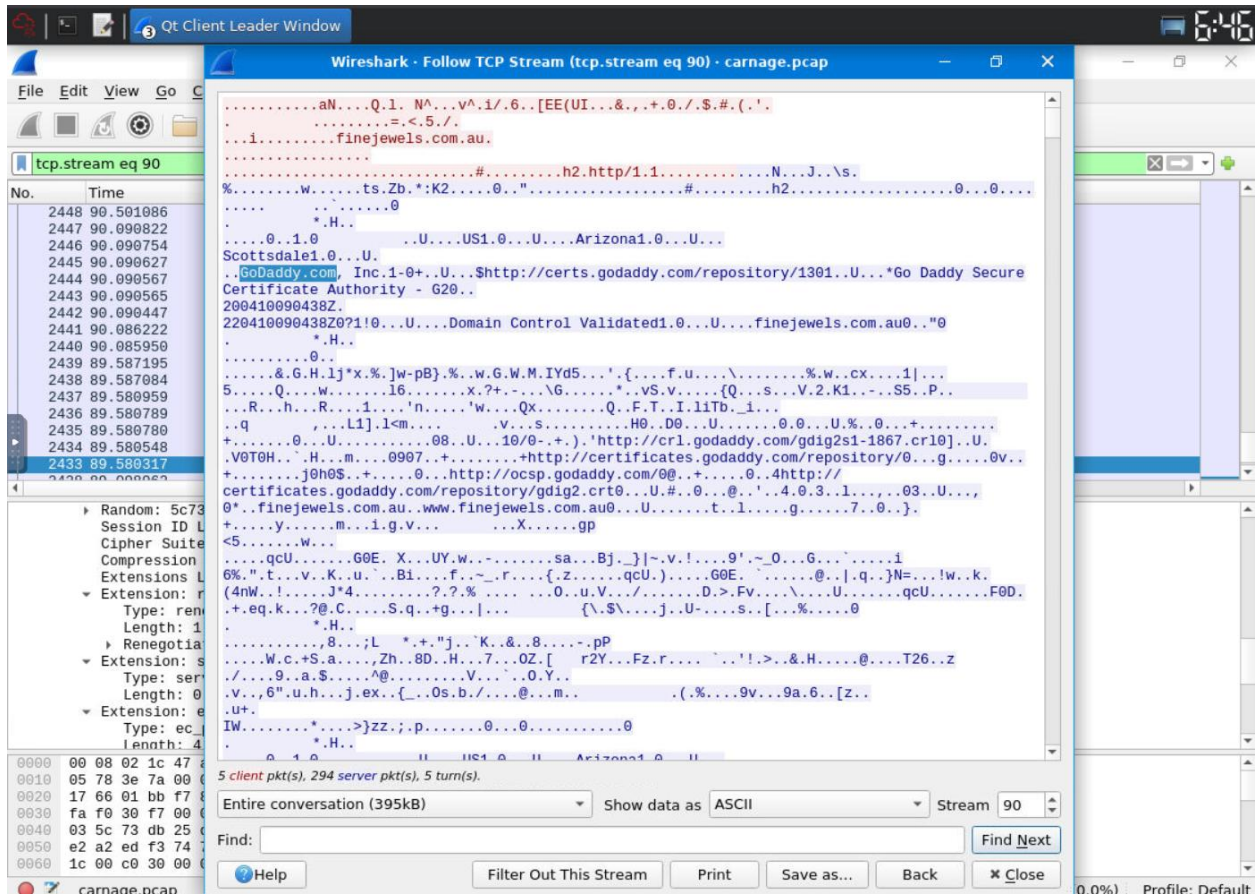
Join our Community and enjoy additional community insights and crowdsourced detections, plus an API key to [automate checks](#).

Security vendors' analysis

CyRadar	Malicious	Fortinet	Malware
Kaspersky	Malware	SOCRadar	Phishing
alphaMountain.ai	Suspicious	Forcepoint ThreatSeeker	Suspicious

Do you want to automate checks?

All three of those domains seemed malicious. The TCP stream of the domains was checked to identify information related to SSL certificates.



The SSL certificate for finejewels.com.au was issued by GoDaddy.

2.3 C2 servers' analysis

The cobalt servers use HTTP/HTTPS (80,8080/443) protocol as per the information from MITRE ATT&CK.

MITRE | ATT&CK

Matrices | Tactics | Techniques | Defenses | CTI | Resources | Benefactors | Blog | Search

SOFTWARE

Cobalt Strike

Cobalt RAT

CoinTicker

Comnie

ComRAT

Concinit1248

Conficker

ConnectWise

Conti

CookieMiner

CORALDECK

CORESHELL

Corona Updates

CosmicDuke

CostaBricks

Covenant

CozyCar

CrackMapExec

CreepyDrive

Token Impersonation: Parent

		.003	Access Token Manipulation: Make and Impersonate Token	Cobalt Strike can make tokens from known credentials. ^[1]
		.004	Access Token Manipulation: Parent PID Spoofing	Cobalt Strike can spawn processes with alternate PIDs. ^{[3][2]}
Enterprise	T1087	.002	Account Discovery: Domain Account	Cobalt Strike can determine if the user on an infected machine is in the admin or domain admin group. ^[4]
Enterprise	T1071	.001	Application Layer Protocol: Web Protocols	Cobalt Strike can use a custom command and control protocol that can be encapsulated in HTTP or HTTPS. All protocols use their standard assigned ports. ^{[1][3][2][4][7]}
		.002	Application Layer Protocol: File Transfer Protocols	Cobalt Strike can conduct peer-to-peer communication over Windows named pipes encapsulated in the SMB protocol. All protocols use their standard assigned ports. ^{[1][5]}
		.004	Application Layer Protocol: DNS	Cobalt Strike can use a custom command and control protocol that can be encapsulated in DNS. All protocols use their standard assigned ports. ^{[1][3][2]}
Enterprise	T1197		BITS Jobs	Cobalt Strike can download a hosted "beacon" payload using BITSAdmin. ^{[6][5][2]}
Enterprise	T1185		Browser Session Hijacking	Cobalt Strike can perform browser pivoting and inject into a user's browser to inherit cookies, authenticated HTTP sessions, and client SSL certificates. ^{[1][2]}
Enterprise	T1059	.001	Command and Scripting Interpreter: PowerShell	Cobalt Strike can execute a payload on a remote host with PowerShell. This technique does not write any data to disk. ^{[1][4]} Cobalt Strike can also use PowerSploit and other scripting frameworks to perform execution. ^{[9][3][5][2]}
		.003	Command and Scripting Interpreter: Windows Command	Cobalt Strike uses a command-line interface to interact with systems. ^{[9][3][2][10]}

From the navigation bar, we were navigated to conversations and then to TCP protocols and the conversation was sorted by the most packets. The addresses with port 80,8080 and 443 were inspected in VirusTotal.

Qt Client Leader Window

6:54

carnage.pcap

Wireshark · Conversations · carnage.pcap

Ethernet · 8		IPv4 · 109		IPv6		TCP · 447		UDP · 256			
Address A	Port A	Address B	Port B	Packets	Bytes	Packets A → B	Bytes A → B	Packets B → A	Bytes B → A	Rel Start	
23.111.114.52	65400	10.9.23.102	63557	18002	15M	12281	15M	5721	753k	884.2113	
10.9.23.102	63555	104.83.84.137	443	9074	10M	1972	107k	7102	10M	880.2065	
10.9.23.102	63465	185.125.204.174	8080	1375	1387k	408	22k	967	1364k	743.6496	
10.9.23.102	63507	185.106.96.158	80	1074	997k	379	20k	695	976k	799.9057	
10.9.23.102	63723	177.149.159.181	25	1069	685k	464	652k	605	33k	1180.5265	
10.9.23.102	63439	136.232.34.70	443	1002	989k	284	16k	718	973k	647.8655	
10.9.23.102	63726	52.97.201.242	25	978	627k	446	592k	532	34k	1186.8759	
10.9.23.102	63610	136.232.34.70	443	976	882k	334	19k	642	862k	1057.1835	
10.9.23.102	63732	52.97.201.210	25	957	616k	441	583k	516	33k	1211.7524	
10.9.23.102	63571	136.232.34.70	443	953	910k	291	16k	662	893k	932.8062	
10.9.23.102	63736	52.97.201.242	25	945	596k	416	562k	529	34k	1219.2012	
10.9.23.102	63738	217.70.178.9	25	933	585k	405	551k	528	34k	1224.0812	
10.9.23.102	63734	62.149.128.200	25	930	574k	397	540k	533	34k	1214.0672	
10.9.23.102	63747	52.98.163.18	25	922	579k	410	543k	512	35k	1241.9027	
10.9.23.102	63752	185.14.56.240	25	900	573k	392	545k	508	27k	1258.1960	
10.9.23.102	63744	52.97.201.194	25	898	574k	402	541k	496	32k	1232.9158	
10.9.23.102	63694	52.97.201.242	25	886	577k	415	544k	471	33k	1145.3025	
10.9.23.102	63749	45.64.187.254	25	883	573k	399	542k	484	31k	1248.3842	
10.9.23.102	63740	52.98.168.178	25	882	585k	412	554k	470	30k	1226.3800	
10.9.23.102	63725	52.97.201.242	25	878	572k	411	541k	467	30k	1186.1673	
10.9.23.102	63737	212.227.15.158	25	877	579k	410	549k	467	29k	1219.9257	
10.9.23.102	63745	193.70.18.144	25	866	581k	416	551k	450	30k	1240.6962	
10.9.23.102	63729	52.97.232.210	25	862	573k	401	543k	461	30k	1201.5996	
10.9.23.102	63730	46.16.61.60	25	861	560k	405	530k	456	30k	1200.2776	

☐ Name resolution
 ☐ Limit to display filter
 ☐ Absolute start time
 Conversation Types ▾

Length: 4

```

0000  00 08 02 1c 47 ae 20 e5 2a b6 93 f1 08 00 45 00  ....G..*....E.
0010  05 78 3e 7a 00 00 80 06 80 80 94 48 c0 ce 0a 09  -x>z.....H..
0020  17 66 01 bb f7 88 e2 ef eb e4 fa df 32 9e 50 18  -f.....2..P..
0030  fa f0 30 f7 00 00 16 03 03 00 4e 02 00 00 4a 03  -0.....N...J..
0040  03 5c 73 db 25 d8 01 c6 ac d9 13 cc f7 77 04 af  -\s-%.....w..
0050  e2 a2 ed f3 74 73 dd 5a 62 10 2a 3a 4b 32 a3 fb  -...ts-Z b-*:K2..
0060  1c 00 c0 30 00 00 22 ff 01 00 01 00 00 00 00 00  -..0..":.....
  
```

carnage.pcap Packets: 70873 · Displayed: 437 (0.6%) · Marked: 1 (0.0%) Profile: Default

185.106.96.158, 185.125.204.174 were found to be the cobalt servers (from the community tab in VirusTotal).

185.106.96.158

drb_ra
3 years ago

Cobalt Strike Server Found

C2: HTTPS @ 185[.]106[.]96[.]158:8888

C2 Server: `survmeter[.]live[.]gscpl[.]R/185[.]106[.]96[.]158[.]gscpl[.]R/`

POST URI: `/supprq/sa/`

Country: United States

ASN: DediPath

Host Header: `ocsp[.]verisign[.]com`

#c2 #cobaltstrike

185.125.204.174

drb_ra
3 years ago

Cobalt Strike Server Found

C2: HTTPS @ 185[.]125[.]204[.]174:4444

C2 Server: `securitybusinpu[.]com/jquery-3[.]3[.]1[.]1[.]min[.]js,185[.]125[.]204[.]174[.]jquery-3[.]3[.]1[.]1[.]min[.]js`

POST URI: `/jquery-3[.]3[.]2[.]1[.]min[.]js`

Country: N/A

ASN: Hydra Communications Ltd

#c2 #cobaltstrike

2.4 Post-infection traffic analysis

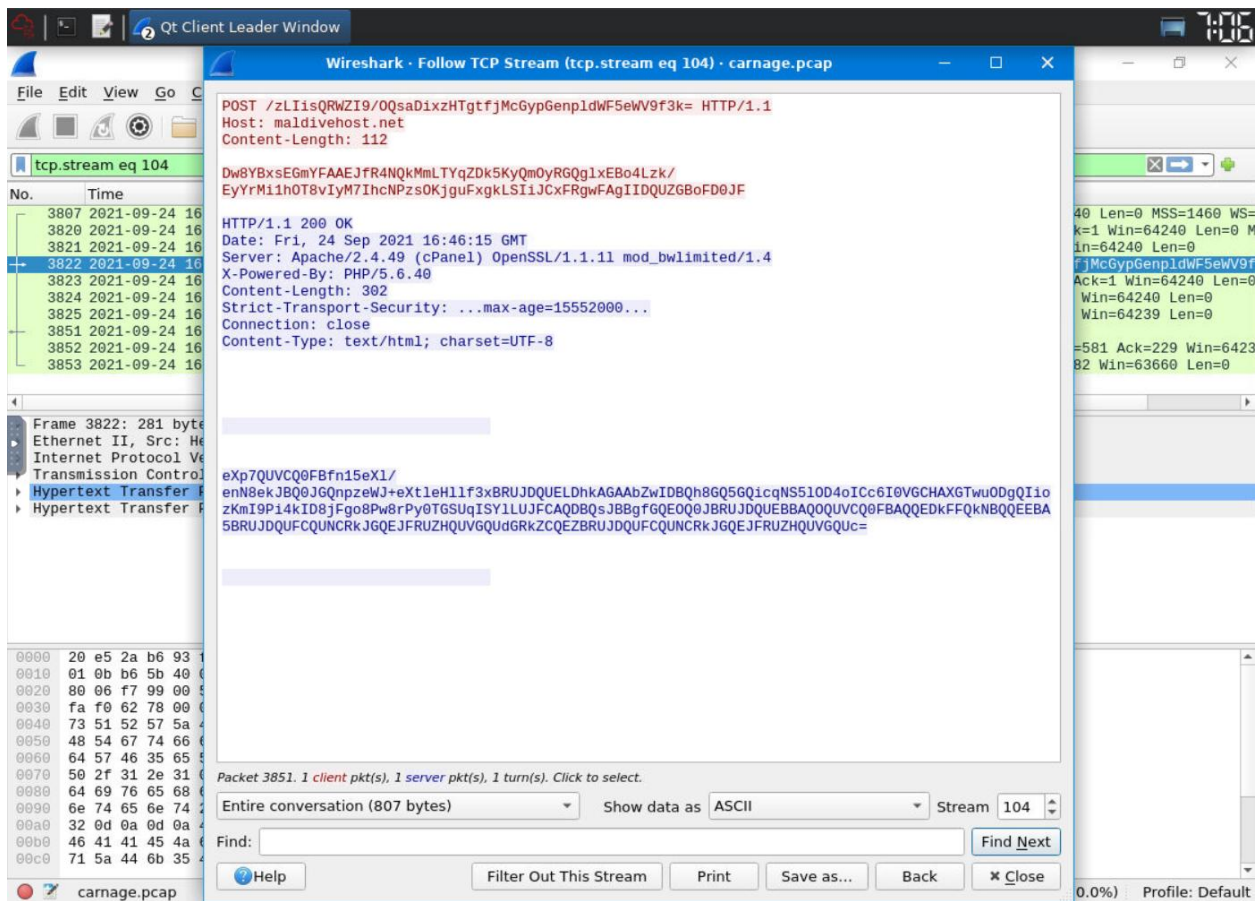
The post-infection traffic was analyzed by filtering for HTTP traffic with POST requests.

The image shows a Wireshark capture of a file named 'carnage.pcap'. The filter bar at the top is set to 'http.request.method == POST'. The packet list shows several HTTP POST requests to the destination IP 208.91.128.6. The selected packet (Frame 13205) is an HTTP POST request. The packet details pane shows the following structure:

- Frame 13205: 826 bytes on wire (6608 bits), 826 bytes captured (6608 bits)
- Ethernet II, Src: HewlettP_1c:47:ae (00:08:02:1c:47:ae), Dst: Netgear_b6:93:f1 (20:e5:2a:b6:93:f1)
- Internet Protocol Version 4, Src: 10.9.23.102, Dst: 185.106.96.158
- Transmission Control Protocol, Src Port: 63533, Dst Port: 80, Seq: 1, Ack: 1, Len: 772
- Hypertext Transfer Protocol

The packet bytes pane shows the raw data of the HTTP POST request, including the status line '200 OK' and various headers.

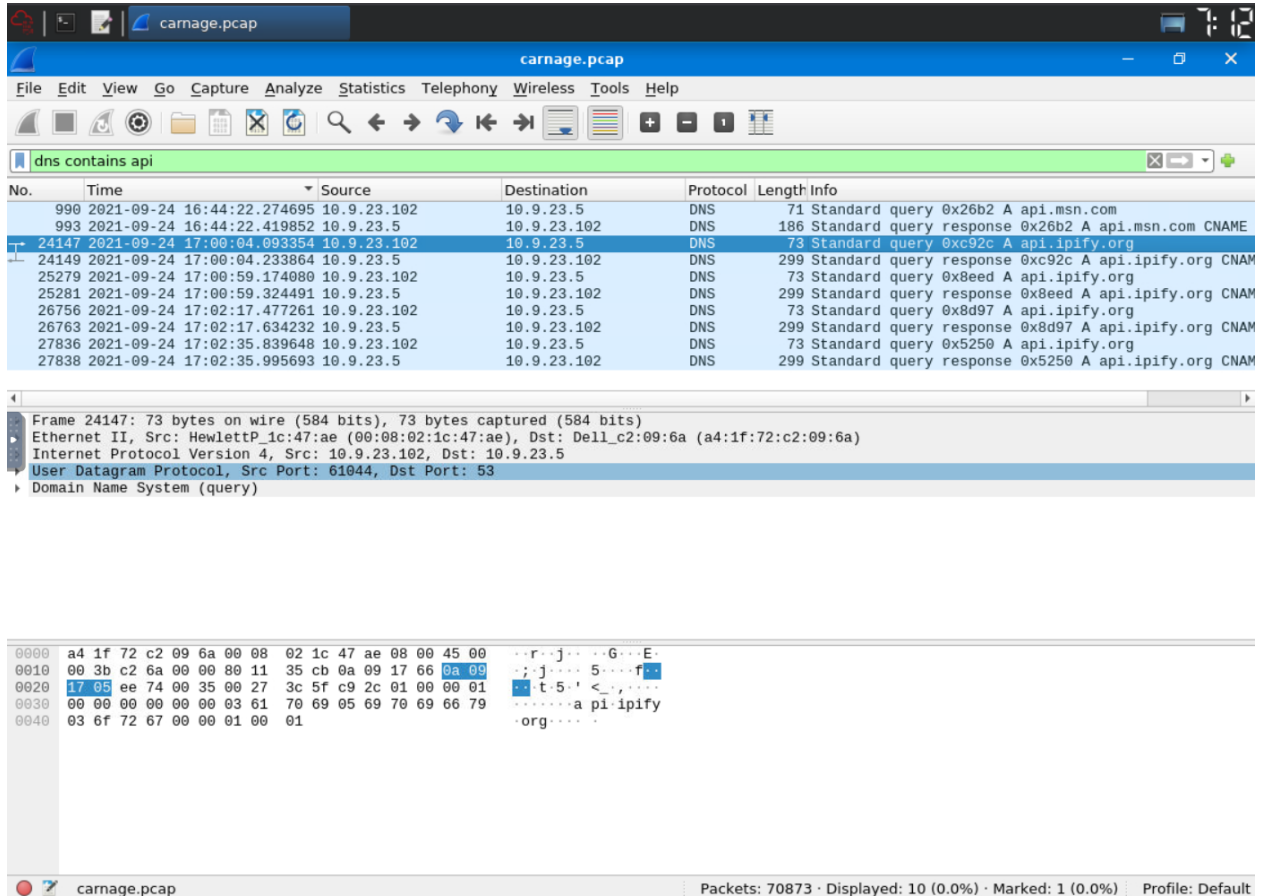
The destination IP address 208.91.128.6 looked suspicious. It was inspected on TCP stream.



So, the host was maldivehost.net. This packet was also the first packet sent out to the C2 server.

2.5 DNS analysis

The malware used api so 'dns contains api' filter was applied. The api.ipify.org seemed to be the api that the malware used.



The image shows a Wireshark capture of a network packet. The top pane displays a list of packets filtered by 'dns contains api'. Packet 24147 is selected, showing a DNS query for 'api.ipify.org'. The middle pane shows the packet details, including Ethernet II, Internet Protocol Version 4, User Datagram Protocol, and Domain Name System (query). The bottom pane shows the raw packet data in hexadecimal and ASCII.

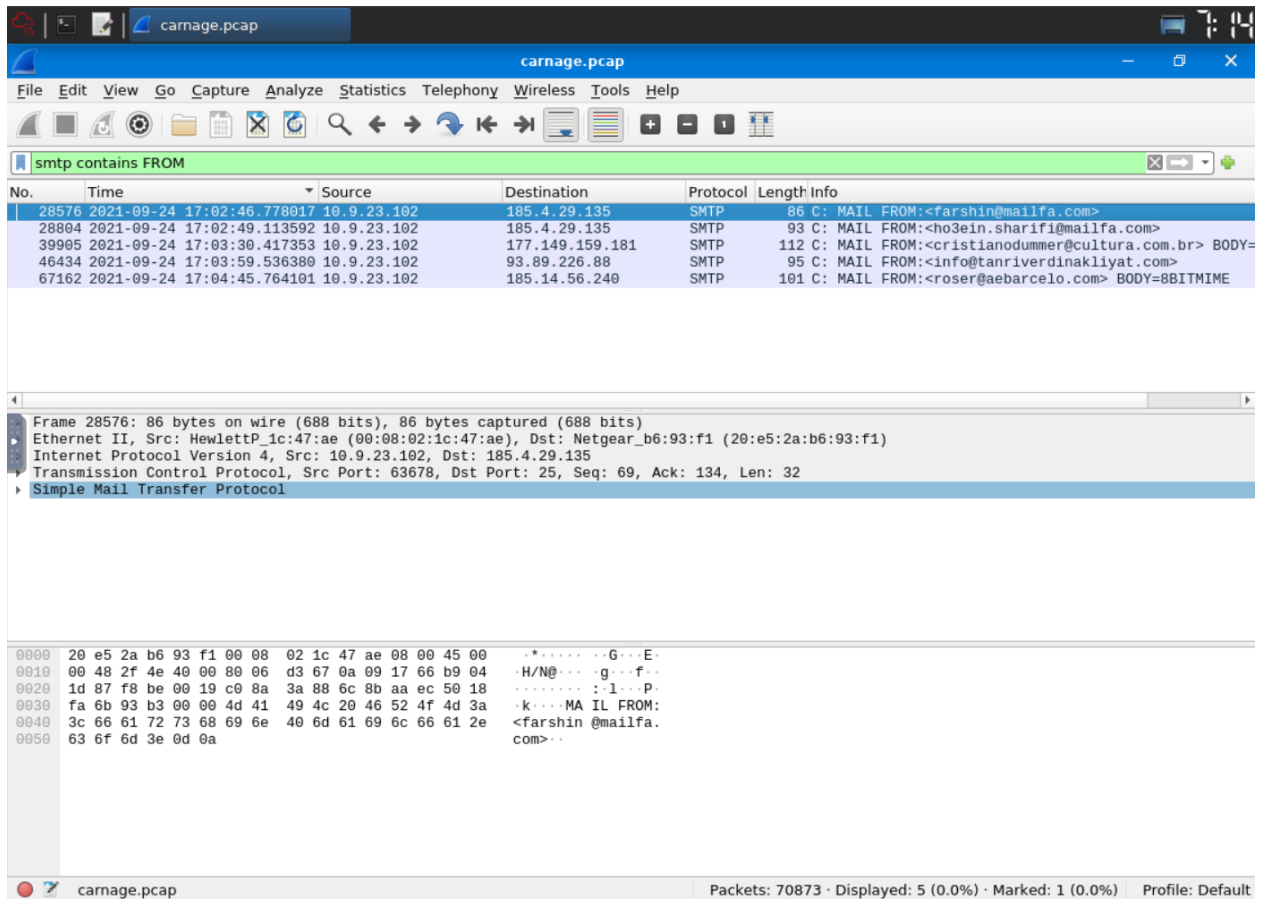
No.	Time	Source	Destination	Protocol	Length	Info
990	2021-09-24 16:44:22.274695	10.9.23.102	10.9.23.5	DNS	71	Standard query 0x26b2 A api.msn.com
993	2021-09-24 16:44:22.419852	10.9.23.5	10.9.23.102	DNS	186	Standard query response 0x26b2 A api.msn.com CNAME
24147	2021-09-24 17:00:04.093354	10.9.23.102	10.9.23.5	DNS	73	Standard query 0xc92c A api.ipify.org
24149	2021-09-24 17:00:04.233864	10.9.23.5	10.9.23.102	DNS	299	Standard query response 0xc92c A api.ipify.org CNAME
25279	2021-09-24 17:00:59.174080	10.9.23.102	10.9.23.5	DNS	73	Standard query 0x8eed A api.ipify.org
25281	2021-09-24 17:00:59.324491	10.9.23.5	10.9.23.102	DNS	299	Standard query response 0x8eed A api.ipify.org CNAME
26756	2021-09-24 17:02:17.477261	10.9.23.102	10.9.23.5	DNS	73	Standard query 0x8d97 A api.ipify.org
26763	2021-09-24 17:02:17.634232	10.9.23.5	10.9.23.102	DNS	299	Standard query response 0x8d97 A api.ipify.org CNAME
27836	2021-09-24 17:02:35.839648	10.9.23.102	10.9.23.5	DNS	73	Standard query 0x5250 A api.ipify.org
27838	2021-09-24 17:02:35.995693	10.9.23.5	10.9.23.102	DNS	299	Standard query response 0x5250 A api.ipify.org CNAME

Frame 24147: 73 bytes on wire (584 bits), 73 bytes captured (584 bits) on interface 0
Ethernet II, Src: HewlettP_1c:47:ae (00:08:02:1c:47:ae), Dst: Dell_c2:09:6a (a4:1f:72:c2:09:6a)
Internet Protocol Version 4, Src: 10.9.23.102, Dst: 10.9.23.5
User Datagram Protocol, Src Port: 61044, Dst Port: 53
Domain Name System (query)

```
0000  a4 1f 72 c2 09 6a 00 08 02 1c 47 ae 08 00 45 00  ..r.j...G...E-
0010  00 3b c2 6a 00 00 00 11 35 cb 0a 09 17 66 0a 09  ..j...5...f.
0020  17 05 ee 74 00 35 00 27 3c 5f c9 2c 01 00 00 01  .t.5' <_,...
0030  00 00 00 00 00 00 03 61 70 69 05 69 70 69 66 79  .....a pi ipify
0040  03 6f 72 67 00 00 01 00 01                          .org...
```

2.6 Mail analysis

The mail can be analyzed by command filter 'smtp contains FROM' which will display SMTP protocols, which is used for emails.



The image shows a Wireshark capture of network traffic from a file named 'carnage.pcap'. The filter 'smtp contains FROM' is applied, displaying a list of SMTP packets. The first packet, No. 28576, is selected and its details are shown in the packet list pane. The packet is an Ethernet II frame from 10.9.23.102 to 185.4.29.135, encapsulating an Internet Protocol Version 4 packet, which in turn encapsulates a Transmission Control Protocol packet (port 63678 to 25), and finally a Simple Mail Transfer Protocol (SMTP) packet. The SMTP packet details show it is a C: MAIL FROM: packet from <farshin@mailfa.com>.

No.	Time	Source	Destination	Protocol	Length	Info
28576	2021-09-24 17:02:46.778017	10.9.23.102	185.4.29.135	SMTP	86	C: MAIL FROM:<farshin@mailfa.com>
28804	2021-09-24 17:02:49.113592	10.9.23.102	185.4.29.135	SMTP	93	C: MAIL FROM:<ho3ein.sharifi@mailfa.com>
39905	2021-09-24 17:03:30.417353	10.9.23.102	177.149.159.181	SMTP	112	C: MAIL FROM:<cristianodummer@cultura.com.br> BODY=
46434	2021-09-24 17:03:59.536380	10.9.23.102	93.89.226.88	SMTP	95	C: MAIL FROM:<info@tanriverdinakliyat.com>
67162	2021-09-24 17:04:45.764101	10.9.23.102	185.14.56.240	SMTP	101	C: MAIL FROM:<roser@aebarcelo.com> BODY=8BITIME

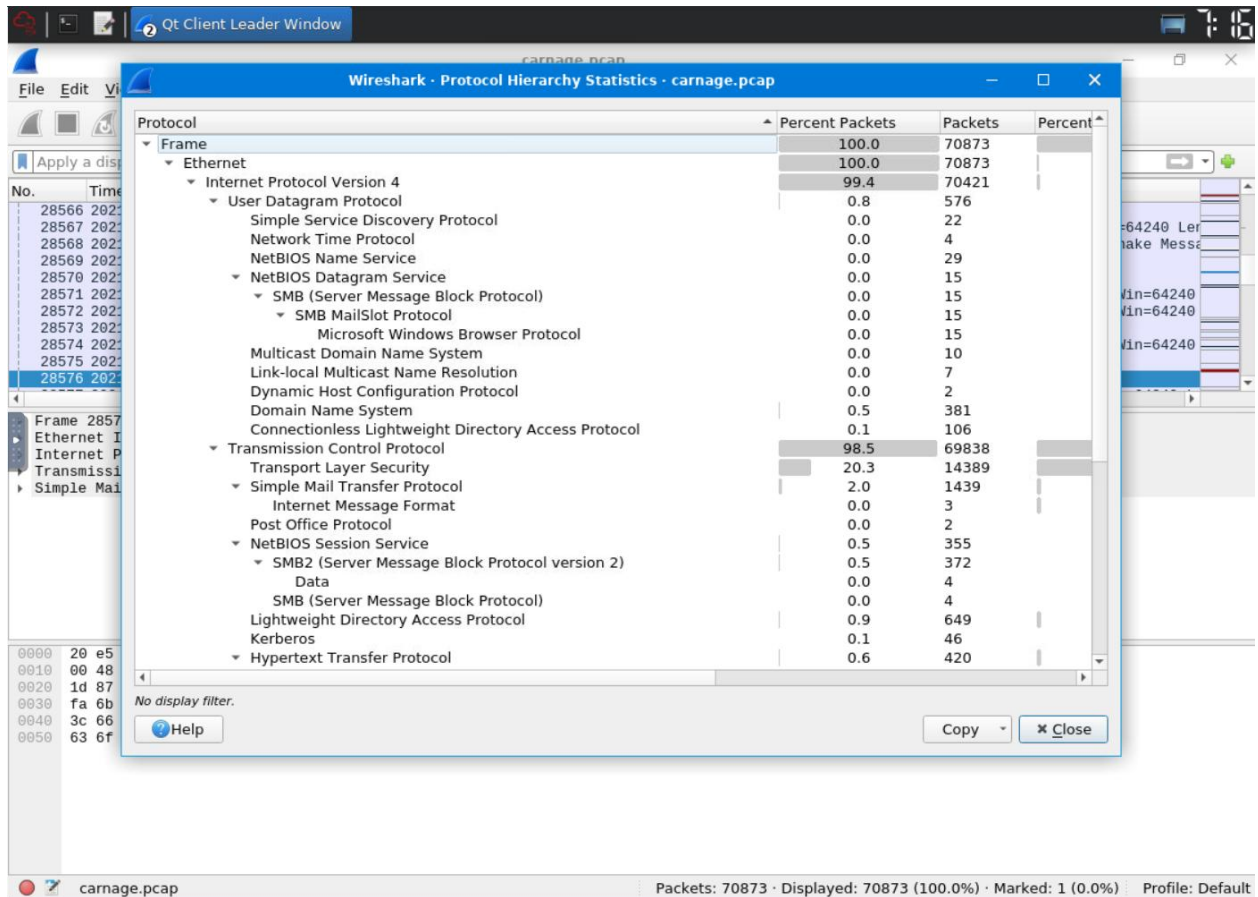
Frame 28576: 86 bytes on wire (688 bits), 86 bytes captured (688 bits)
Ethernet II, Src: HewlettP_1c:47:ae (00:08:02:1c:47:ae), Dst: Netgear_b6:93:f1 (20:e5:2a:b6:93:f1)
Internet Protocol Version 4, Src: 10.9.23.102, Dst: 185.4.29.135
Transmission Control Protocol, Src Port: 63678, Dst Port: 25, Seq: 69, Ack: 134, Len: 32
Simple Mail Transfer Protocol

```
0000  20 e5 2a b6 93 f1 00 08 02 1c 47 ae 08 00 45 00  .*. . . . .G. . .E.  
0010  00 48 2f 4e 40 00 80 06 d3 67 0a 09 17 66 b9 04  .H/N. . . .g. .f.  
0020  1d 87 f8 be 00 19 c0 8a 3a 88 6c 8b aa ec 50 18  . . . . .:l. .P.  
0030  fa 6b 93 b3 00 00 4d 41 49 4c 20 46 52 4f 4d 3a  .k. . .MA IL FROM:  
0040  3c 66 61 72 73 68 69 6e 40 6d 61 69 6c 66 61 2e  <farshin@mailfa.  
0050  63 6f 6d 3e 0d 0a                                com> . .
```

The first mail came from farshin@mailfa.com.

2.7 Protocol-packets analysis

From the statistics, we were navigated to protocol hierarchy.



There is information on protocols and the number of packets they use. SMTP used 1439 packets.

3 Findings

This table answers the challenge questions from the room.

1. The victim's host established its first HTTP connection to a malicious IP on 2021-09-24 at 16:44:38.
2. A malicious archive named documents.zip was downloaded by the client via unencrypted HTTP. The archive contained a file named chart-1530076591.xls
3. The zip file was hosted on the domain attirenepal.com.
4. The web server hosting the zip file was identified as LiteSpeed, running version PHP/7.2.34.
5. Malicious files were additionally downloaded from three suspicious domains: finejewels.com.au, thietbiagt.com, and new.americold.com.
6. The SSL certificate for finejewels.com.au was issued by GoDaddy.
7. The malware utilized a DNS query to the domain api.ipify.org on 2021-09-24 at 17:00:04 to check the victim's IP address.
8. Two Cobalt Strike C2 servers were identified at IP addresses 185.106.96.158 and 185.125.204.174.
9. The first C2 server (185.106.96.158) resolved to the domain survmeter.live and utilized the Host header ocsp.verisign.com.
10. The second C2 server (185.125.204.174) resolved to the domain securitybusinpuff.com.
11. Post-infection traffic was directed to the domain maldivehost.net.
12. The Server header for the post-infection domain was identified as Apache/2.4.49 (cPanel) OpenSSL/1.1.1l mod_bwlimited/1.4.
13. The initial packet sent out to the C2 server had a length of 281. The first eleven characters sent from the victim host to the malicious post-infection domain were zLLisQRWZI9.
14. Malicious spam (malspam) activity was detected in the traffic analysis.
15. The first observed sender address (MAIL FROM) in the traffic was farshin@mailfa.com.
16. A total of 1439 packets were associated with SMTP traffic.